

Development of a Tangible Interactive Playware for Autism: Through the Lens of Design Science

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Abstract

This paper introduces the application of a novel methodology to develop iTILES, a tangible interactive playware system designed to provide tailored interventions for children with autism spectrum disorder (ASD). Using the design science research (DSR) methodology, we identify meta-requirements, develop a prototype integrating hardware, software, and games. iTILES provides customizable, multisensory, and data-driven therapeutic tools, offering engaging and adaptive learning experiences, and formulate theory-driven design principles. Key contributions include advancing the design of interactive playware for ASD therapy, creating four structured design principles, and enhancing theoretical insights into technology-assisted interventions for developmental disorders. This lays the groundwork for future studies on deploying the artifact and advancing a design theory for tangible interactive playware.

CCS Concepts

• **Human-centered computing** → *Accessibility design and evaluation methods.*

Keywords

Tangible Interactive Playware, Playware for Autism, Children with ASD, Design Science Research, Design Principles

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1 Introduction

Individuals with autism spectrum disorder (ASD) face significant challenges in their capacity to engage and communicate with others, as well as in their comprehension of the environment around them. This condition is characterized by a broad spectrum of behavioral expressions, showing vast variability across individuals [36]. Children with ASD are among the most rapidly increasing developmental disabilities. Recent data reveals that the prevalence of autism in children has reached 1 in 36 in the US [20], with similar patterns observed in other parts of the world [27]. The growing prevalence of autism underscores the critical need for effective interventions and support mechanisms. Early and tailored interventions can significantly enhance their developmental outcomes, improving communication skills, social interactions, and overall quality of life.

To improve the quality of life for those with ASD, there has been an increasing emphasis on employing digital aids [5]. Among various approaches, tangible interactive playwares are widely used

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to support children with ASD [38]. Research has demonstrated that it enhances cognitive development in children with ASD by improving their attention, play, and gaming abilities [11, 38]. These platforms also boost motor skills, fostering an engaging and exploratory learning environment [4, 15], and offering health benefits such as improved mobility and self-esteem [3]. Our system builds on these insights, aiming to create a digital tool to support the development of children with ASD. It allows therapists to customize games that encourage active participation, focusing on enhancing cognitive and motor abilities, and fostering interpersonal interactions [32].

Existing studies have designed tangible interfaces to enhance cognitive or motor abilities as individual skills. Co-design and user-centered methods, such as participatory design, have been employed to create artifacts. Although these approaches engage various stakeholders in the design process, they do not offer a structured framework to develop design principles [9, 30, 33]. There is a lack of research on identifying design principles that enable designing tangible interactive playware from early conceptualization to production-level deployment. There is also limited research on theory-driven approaches to designing such artifacts.

To address this, we adopt design science research (DSR) methodology, following established DSR principles and guidelines in information systems [14, 23]. We define our problem, identify meta-requirements (MRs) and formulate theory-driven design principles (DPs). These principles are instantiated in our prototype iTILES, a tangible interactive playware for children with ASD. This short paper presents the initial cycle of the DSR process.

The contributions of this study include: first, identifying meta-requirements for developing a tangible interactive playware using DSR; second, iteratively developing a deployable prototype that caters to the needs of children with ASD; third, we synthesize the design and development with the stakeholders into four design principles.

2 Related Work

Technology has been shown to be effective in managing stress and anxiety among people with ASD [5]. Prior literature shows that children with ASD find interactive systems engaging and reliable, and therefore numerous researchers have developed interactive technology-based interventions [2, 12]. A specific technology-based intervention is the tangible interface. Tangible interfaces facilitate learning by stimulating all senses, supporting the comprehensive development of children, including those with physical or cognitive disabilities. Prior literature highlights their benefits in enhancing cognitive, motor, and social skills. For instance, M. A. Carreño-León et al., [7] developed an interactive system to support cognitive abilities in children with ASD. Similarly, Al Mahmud & Soysa [1] and Alessandrini [2] designed cost-effective, therapy-assisting systems for the same demographic. However, these studies often lack a systematic approach to derive design principles.

Another thread of studies that includes tangible interfaces focuses on motor abilities. Research indicates that the use of tangible interfaces positively impacts children's learning by promoting development through play. Games have been used frequently as beneficial strategies in the treatment of ASD [7]. Games, particularly those designed for children, reinforce motor skills [15]. This

highlights the opportunities in integrating games with tangible interfaces in the form of tangible interactive playware. Studies show that training with games using tangible interfaces can improve physical abilities such as balance, mobility, and agility [17]. These interfaces enhance cooperative and associative play sequences in children with ASD, reducing the duration and frequency of solitary activities [35]. Tangible interfaces have also been developed to help children with ASD improve individual skills, such as fine motor skills [12]. Lastly, a separate line of research emphasizes social abilities, which our study does not explore [12, 39].

Existing research covers various aspects of cognitive, motor and social skills, yet there is a notable gap in the existence of design guidelines for tangible interactive playware. A few studies provide such design guidelines; for instance, Mehta et al. [22] developed an inclusive educational game but the focus was not on ASD. Bouayad et al. [6] proposed a personalized voice alert system to improve user compliance and reduce wandering among individuals with ASD. Additionally, Bondioli et al. [5] developed a digital tool to help reduce anxiety of children with ASD. Although these studies employ design science, they do not develop a tangible interactive playware.

While prior studies in HCI have explored the use of tangible interfaces, they often lack a theory-driven approach to derive design principles. Our study addresses this gap by proposing design principles developed through a structured framework, specifically for tangible interactive playware.

3 Methodology

We adopt a DSR approach to develop tangible interactive playwares [23]. DSR focuses on developing and evaluating innovative solutions to complex, real-world issues [14, 23]. The DSR process is divided into the following steps: defining the problem and solution space, designing and developing, demonstrating, evaluating, and communicating [19, 37]. In this short paper, we define the problem and present the ongoing design and development of our prototype.

3.1 Defining the Problem Space and the Solution Space

The development of our artifacts for children with ASD is predicated on understanding the nuanced challenges children face in engaging with their environments and learning new skills. The problem objectives for developing an artifact (tangible interactive playware) can be described by defining the problem space and solution space [16, 37]. In the problem space we define the needs, goals, requirements and stakeholders (Table 1) to address the solution developed using the DSR approach [19].

3.2 Defining the Objective of the Solution Space through Meta-Requirements

Meta-requirements outline the objectives tackled within a specific class of problems and are critical elements of a design theory [24]. For developing our artifact, the following meta-requirements were derived from the needs of the system's stakeholders (three therapists, one designer, four software developers and one hardware developer). The needs were gathered through semi-structured interviews with the stakeholders.

Table 1: Defining the Problem Space

Elements	Description
Stakeholders	1. Designers, software and hardware developers, therapists and children with ASD.
Needs	Technologies to enhance cognitive, motor, social skills through: 1. Enhanced Sensory Interaction offering structured, predictable interactions [31]. 2. Customizable Learning Paths providing the ability to customize interactions according to individual needs [10]. 3. Data-Driven Insights to facilitate therapists in understanding and improving therapeutic interventions [28].
Goals	1. Engagement and Usability to create an intuitive and engaging interactive system that encourages regular use and facilitates ease of interaction without overwhelming the user [18, 29]. 2. Therapeutic and Educational Value to leverage the interactive capabilities through games to support both cognitive and motor skill development [26, 29]. 3. Stakeholder Support to equip therapists with tools to monitor progress, make informed decisions and tailor interventions based on real-time data [18].
Requirements	1. The artifact, combining hardware, software and game elements would be designed to meet essential technical and functional requirements [26]. 2. The hardware would offer multimodal sensory feedback and wireless connectivity for flexible integration. 3. The software would include a user-friendly application and analytics dashboard that enables therapists to customize, manage, and analyze educational activities. 4. The games would be specifically designed for children with ASD, aimed at improving cognitive and motor skills through interactive play.

- (1) *Multimodal Interaction Capability (MR1)* - It is important to create engaging sensory experiences that capture and hold the attention of children with ASD, thereby enhancing their interaction with the system. To ensure this, the artifact should provide a sensory-rich interaction environment, support haptic, visual, and auditory feedback. When speaking to therapists about the system design, they asked “could I adjust the pressure sensitivity? Could I control which tile lights up, or which one vibrates?” These features allow therapists to engage with children with ASD in diverse ways and accommodate their varied sensory processing preferences.
- (2) *Adaptive Learning and Customizability (MR2)* - Adaptive learning should be central to personalizing the therapeutic and educational experiences of each individual using the artifact. This adaptability allows therapists to adjust difficulty levels and activity types based on real-time user feedback and performance data. Regarding the customizability and development of the games, the designers noted, “The therapists affirmed a lot of skills, they also talked about specifics for each student, they looked at skills that they wanted to develop, how they fit between each student to fit under certain groups of skills, such as personalities. That’s where we came up with different levels and such, so that the therapist can decide which level aligns with the student and then apply that.”
- (3) *Data-driven Insights (MR3)* - Data-driven insights should be a fundamental part of the artifact for providing evidence-based feedback to therapists and educators. This meta-requirement emphasizes the importance of an analytics dashboard that collects and displays interaction data. The developers conveyed the therapists’ needs, “We considered what data we

could collect from the tiles for the therapists, how they wanted it to appear, how they could access it later in the games, and how they could use it to monitor the progress of their students.” This capability allows therapists to make informed decisions about how to adjust therapy sessions and educational activities for better outcomes. Therapists noted that “sessions are usually back to back, so progress notes are just highlights” and that it can be “difficult to do a progress note for each individual student. It’s usually what we remember, we can’t take quantitative data.” Quantitative insights gathered from user interactions would greatly enhance therapists’ ability to track progress, and identify areas needing more focus.

- (4) *Accessibility and Usability (MR4)* - This ensures that the artifact is user-friendly in the physical design and digital interfaces, easy to set-up, and requires little to no training. This requirement is particularly important in diverse therapeutic settings where users’ technical proficiency can vary widely. Some features that designers need to consider include: sensors, size, and weight of hardware, user experience of the software application, and game design flows. For example, to address the usability aspect of the sensing mechanism, designers evaluated different sensors to accommodate the varying pressure levels exerted by diverse users on the tiles. As one designer working on the iTILES explained, “We’ve explored the use of various pressure sensors mounted on the base of the floor tiles. So when someone stands on it, the pressure sensors measure that the tiles have been stepped on.”

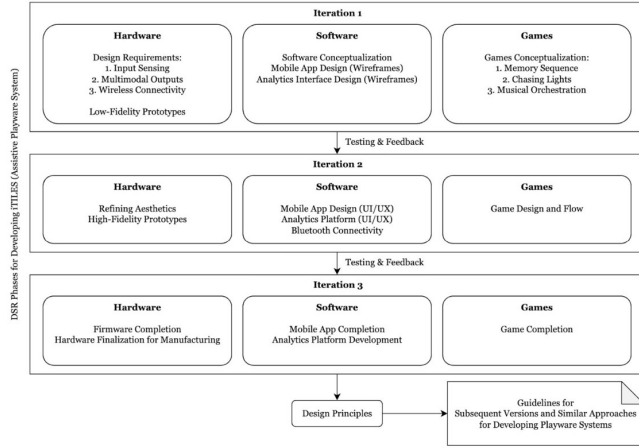


Figure 1: DSR Iterations for Developing iTILES.

With these meta-requirements defined, the subsequent sections will detail the iterative design process that translates these meta-requirements into tangible system components (Figure 1).

3.3 DSR Iterations for Developing iTILES

The development process of iTILES was iterative (Figure 1), allowing continuous refinement based on theoretical insights, internal feedback, and testing. It involved various components (hardware, software, and games) being individually developed in phases to contribute to the design of the overall playware system. Each iteration aimed to enhance the system’s functionality, usability, and effectiveness in therapeutic settings for children with ASD as depicted in Figure 1.

3.3.1 Iteration 1: Initial Conceptualization. The first iteration lasted six months and was exploratory, involving three therapists, one designer and five developers (four software, one hardware) to establish design objectives for hardware, games, and software.

Hardware: The first hardware iteration involved designing and 3D printing prototypes with key functionalities: Input Sensing for touch, tilt, and connection recognition; Multimodal output using RGB LEDs, haptics, and audio for interaction; and Bluetooth for wireless connectivity with the software platform. Various tile shapes and sizes were considered (Figure 2a), with the hexagonal shape taking precedence to create a modular system (Figure 2c).

Software: The initial framework for app connectivity and tile-software integration was established, alongside plans for an analytics dashboard to manage the playware system. We also gathered requirements for the dashboard through interviews with therapists.

Games: The games initially conceptualized and designed during this iteration aimed to utilize the interactive potential of the hardware. We focused on the development of the initial design of games like "Memory Sequence," "Chasing Lights," and "Musical Orchestration," aiming to enhance cognitive skills and motor skills.

3.3.2 Iteration 2: Prototype Development. In this iteration we further improved the hardware, software, and game components. This iteration took another six months.

Hardware: During this phase, the focus shifted towards developing high-fidelity prototypes. This included refining the aesthetics, selecting materials, and considering the mechanical design and

connectivity aspects essential for product realization. Feedback on different vibration patterns and other interactive elements was sought from external designers, the end-user, to refine the design further.

Software: A mobile app was developed, addressing challenges such as Bluetooth connectivity issues and integrating the games into the app’s user flow. The architecture of the system was adjusted leading to separation of the games and analytics into distinct platforms, with games housed in the app (Figure 3a) and analytics accessible via a web-page (Figure 3b). This was a significant strategic decision to streamline user experience and focus on each component’s specific requirements.

Games: Three games were developed as therapeutic interventions: (1) Memory Sequence, aimed to enhance memory by requiring users to replicate light patterns; (2) Chasing Lights, focused on improving motor skills by having users tap sequentially lighting tiles; and (3) Musical Orchestration, designed to foster collaboration by producing sounds when tiles are pressed or shaken.

3.3.3 Iteration 3: Refinement and finalization of the playware. This iteration further looked into refinement, testing, and preparations to conduct user studies, which aimed to prepare the system for deployment. The third iteration spans approximately four months and is presently ongoing.

Hardware: In this iteration, the hardware was finalized with integrated functionalities and firmware, achieving a production-ready quality (Figure 3d). The design of the circuit board and firmware was completed, refining the hardware components to ensure reliable performance in user environments.

Software: The focus was on the user interface (UI) and user experience (UX) to ensure consistency across the system. The code quality of the software was improved, and the analytics dashboard was refined (Figure 3b). Efforts were made to polish the UI and develop a backend system capable of processing raw data, ensuring that the user interfaces were consistent with each other.

Games: The games were largely completed by this stage and underwent additional testing by developers and therapists to ensure smooth gameplay.

4 Design Principles

Our preliminary design principles are deduced from a thematic analysis of the literature, meta-requirements, and design iterations [24]. Following the schema provided by [13], we developed our design principles to include the aims (A1 - A4) of the principle, mechanisms (M1 - M12) used to achieve the aim, and a rationale that justifies the principle. The specific aim and mechanisms are detailed within each design principle described below.

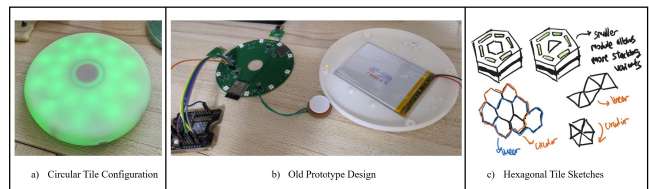


Figure 2: Early Prototypes and Sketches

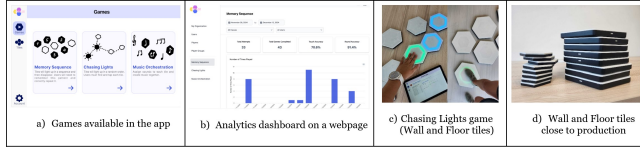


Figure 3: Software Application, Analytics Dashboard and Tangible Interfaces

DP1: Principle of Multimodal Engagement - Research indicates that multisensory environments significantly boost engagement and focus in children with ASD. Studies have shown that tactile, auditory, and visual stimuli enhance cognitive and motor skill development [1], supported by findings from [15] and [4]. Therefore, creating adaptable sensory experiences for children with ASD leads to more effective learning. This was aimed through Foster Sensory Engagement and Cognitive Stimulation (A1) (as specified by MR1) to leverage the interplay between sensory input and cognitive processing to enhance learning and therapeutic outcomes for children with ASD. The aim is implemented through three mechanisms: (1) Multisensory Stimulation (M1) - Harness the engaging potential of multisensory experiences to capture and maintain interest and attention, essential for learning and therapy; (2) Feedback Loop (M2) - Create immediate and intuitive feedback systems that validate user actions, reinforcing learning through a cycle of action and response; (3) Sensory Adaptation (M3) - Provide adaptable sensory experiences to individual sensitivities and preferences environment.

DP2: Principle of Adaptive and Customizable Interactive Experience - Shaer & Hornecker [25] emphasize the benefits of customizable systems for users' specific needs (as specified by MR2). Similarly, [21] and [11] highlight how game design autonomy lets ASD players learn at their own pace, enhancing engagement. This design principle through the support of autonomy and personalized learning (A2) aims to enable a sense of control and self-directed learning which is known to increase motivation and engagement. The aim is implemented through: (1) Adjustable Complexity (M4) - Allow the complexity of tasks to be modified to suit the users' abilities, encouraging continued engagement through achievable challenges; (2) Individualized Experience (M5) - Facilitate a system that can be tailored to each user's specific therapeutic and educational goals, ensuring a relevant and impactful experience; (3) User-Controlled Pacing (M6) - Provide users with the ability to influence the pace of their interaction, promoting a sense of autonomy and comfort with the learning process.

DP3: Principle of Data-Driven Insights and Analytics - Tangible interfaces provide rich data on user interactions, such as touch, movement, and object manipulation. This data can be used to understand user behavior patterns, preferences, and engagement levels in real-time and inform the customization of therapy and learning experiences [1]. This is facilitated through Informed Decision-Making with Actionable Data (A3) to collect and analyze data from interactions (as outlined in MR3), providing insights that inform therapy and learning strategies. The aim is implemented through: (1) Data Collection (M7) - Systematically gather data from user interactions to build a comprehensive understanding of user behavior and progress; (2) Insightful Analysis (M8) - Analyze interaction data to extract patterns and insights that can inform the

customization of therapy and learning experiences; (3) Continuous Improvement (M9) - Use data to drive iterative improvements in both the playware system and therapeutic interventions.

DP4: Principle of Accessibility and User-Centricity - Tangible interfaces naturally afford simplicity and direct manipulation, which can reduce cognitive load and make systems more intuitive [8]. The physicality of tangible interfaces can make the learning process more intuitive, allowing users to grasp system functionalities quicker and with less instruction, reducing time and effort required for users to become proficient (as specified in MR4). This is aimed by Ensuring Accessibility and Intuitive Use (A4) to make the system easily accessible and intuitive for all users, regardless of their technical expertise or cognitive abilities. This is aimed through: (1) Simplification (M10) - Reduce complexity in system interfaces to facilitate ease of use and lower the barrier to entry for all users; (2) Universal Access (M11) - Adopt design strategies that cater to a broad user base, taking into account varying abilities and preferences; (3) Minimized Learning Curve (M12) - Ensure that users can quickly become proficient with the system, allowing them to focus on therapeutic and learning outcomes.

5 Future Work

As the design principles generated in this study for this artifact are preliminary, one key future work is validation which would be performed using framework for evaluation in design science research (FEDS) [34]. We plan to deploy iTILES across three schools and pursue the following steps to validate our design principles: (1) *Explicate the goals*: We will formulate hypotheses to check if the artifact improves cognitive and motor abilities and conduct naturalistic evaluation at the schools, supervised by therapists; (2) *Choose a strategy or strategies for the evaluation*: Quantitative and qualitative methods would be used to gather data. Quantitative methods would include pre- and post-intervention assessments of cognitive and motor skills. Qualitative feedback would be collected through interviews and surveys with therapists and observations of children's engagement and interaction with the playware; (3) *Determine the properties to evaluate*: Based on the evaluation, we would assess the playware for specific areas of improvement, such as game difficulty levels, appropriate interface, and sensitivity adjustments; (4) *Design the individual evaluation episodes*: This might involve software updates, hardware modifications, or alterations to the therapeutic activities incorporated into playware.

Through iTILES, we aim to provide a comprehensive architecture that integrates hardware, software, and game elements to create a tangible interactive playware. Furthermore, the development of iTILES contributes to the theoretical foundations and literature of tangible interactive playware in therapeutic settings. Finally, our design principles and the use of DSR provides insights to guide future research and design of similar systems in technology-assisted therapies for children with developmental disorders.

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